

Comparative Analysis of Deep Learning Models for Real-World ISP Network Traffic Forecasting

面向真实世界互联网服务提供商网络流量预测的深度学习模型比较分析

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Abstract—Accurate network traffic forecasting is crucial for Internet service providers to optimize resources, improve user experience, and detect anomalies. Until recently, the lack of large-scale, real-world datasets limited the fair evaluation of forecasting methods. The newly released CESNET-TimeSeries24 dataset addresses this gap by providing multivariate traffic data from thousands of devices over 40 weeks at multiple aggregation granularities and hierarchy levels. In this study, we leverage the CESNET-TimeSeries24 dataset to conduct a systematic evaluation of state-of-the-art deep learning models and provide practical insights. Moreover, our analysis reveals trade-offs between prediction accuracy and computational efficiency across different levels of granularity. Beyond model comparison, we establish a transparent and reproducible benchmarking framework, releasing source code and experiments to encourage standardized evaluation and accelerate progress in network traffic forecasting research.

摘要——精准的网络流量预测对于互联网服务提供商优化资源利用、提升用户体验及检测异常情况至关重要。长期以来，缺乏大规模真实世界数据集限制了对预测方法的客观评估。新发布的 CESNET -TimeSeries24数据集通过提供涵盖数千台设备、持续40周、包含多种聚合粒度和层级结构的多变量流量数据，有效填补了这一空白。本研究利用 CESNET -TimeSeries24数据集对最先进的深度学习模型进行系统性评估，并提出实用见解。分析结果进一步揭示了不同粒度水平下预测精度与计算效率之间的权衡关系。除模型比较外，我们还建立了透明且可复现的基准测试框架，并公开源代码及实验数据，以推动网络流量预测研究领域的标准化评估并加速技术进步。

Index Terms—Neural networks, deep learning, network traffic forecasting, network traffic prediction, network monitoring.

索引术语—神经网络、深度学习、网络流量预测、网络流量预判、网络监控。

I. INTRODUCTION

I. 引言

TRAFFIC monitoring is a cornerstone of effective network management and cybersecurity, providing Internet Service Providers (ISPs) with critical insights to detect anomalies, mitigate congestion, and maintain network performance [1]. The surge in video streaming, cloud computing, and online gaming is driving rapid growth in Internet usage, contributing to increasingly complex and less predictable network traffic. Efficient network monitoring allows 流量监控是高效网络管理和网络安全的基石，为互联网服务提供商（ISP）提供关键洞察，以检测异常情况、缓解网络拥塞并维持网络性能[1]。视频流媒体、云计算和在线游戏的迅猛发展推动了互联网使用量的快速增长，导致网络流量日益复杂且难以预测。高效的网络监控能够

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ISPs to maintain service quality, mitigate security risks, and optimize bandwidth in real time [2].

互联网服务提供商（ISP）需实时维护服务质量、降低安全风险并优化带宽[2]。

However, real-time monitoring alone is insufficient for proactively managing network resources. To anticipate variations in demand and prevent service disruptions, ISPs increasingly adopt advanced forecasting techniques to predict traffic patterns, optimize resource allocation, scale network capacity, and sustain service quality under fluctuating loads [3]. The predictions can be integrated into Software-Defined Networking (SDN), Network Function Virtualization (NFV), and Service Level Agreement (SLA) frameworks [4], [5], [6], [7], allowing for programmable orchestration of network resources based on predicted traffic demands. For example, SDN controllers can leverage traffic forecasts to proactively adjust routing paths and traffic engineering policies, minimizing congestion and improving latency [4]. NFV environments benefit from forecasting by dynamically scaling Virtual Network Functions (VNFs) and optimally placing them across distributed cloud and edge data centers, thus enhancing service elasticity and reducing operational costs [5]. SLA monitoring benefits from predicting violations early and guiding automated resource allocation, which improves network performance [6], [7].

然而，仅依靠实时监控不足以主动管理网络资源。为预测需求波动并防止服务中断，互联网服务提供商日益采用先进的预测技术来分析流量模式、优化资源配置、扩展网络容量，并在负载波动情况下保障服务质量[3]。这些预测结果可整合至软件定义网络（SDN）、网络功能虚拟化（NFV）及服务等级协议（SLA）框架中[4][5]。[6], [7]该机制支持根据预测的流量需求对网络资源进行可编程化调度。例如，SDN 控制器可利用流量预测数据主动调整路由路径和流量管理策略，从而降低网络拥塞并提升延迟性能[4]；NFV 环境通过动态扩展虚拟网络功能（VNF）并将其最优部署于分布式云中心与边缘数据中心，充分发挥预测分析优势，显著增强服务弹性并降低运营成本[5]；SLA监控则能通过早期识别违规行为并指导自动化资源分配，有效提升网络性能[6]。[7]。

The rise of diverse, high-bandwidth services has significantly increased network traffic variability. Traditional models like ARIMA and exponential smoothing, which assume linearity, struggle with ISP data due to prevalent non-linear and high-frequency fluctuations, especially during peak traffic hours [8].

多样化、高带宽服务的兴起显著增加了网络流量的波动性。基于线性假设的传统模型（如 ARIMA 法和指数平滑法）难以准确处理互联网服务提供商的数据，因为这些数据普遍存在非线性和高频波动特征，尤其是在流量高峰时段[8]。

The adoption of deep learning models, particularly neural networks, has been driven by their ability to capture complex temporal dependencies across forecasting domains [9]. This capability significantly enhances network traffic forecasting and provides ISPs with deeper insights into usage trends and anomalies. However, Wu and Keogh [9] emphasize that evaluating recent advancements in forecasting methods remains difficult due to the lack of long-term real-world datasets. They show that relying on synthetic datasets can create the illusion of progress that does not generalize to real-world conditions. Similarly, Ferreira et al. [10] identify the absence of a reference dataset for network traffic forecasting as a major obstacle to performance evaluation. They note that most approaches are still tested only on artificially generated data or non-publicly available data.

深度学习模型（尤其是神经网络）的广泛应用，主要得益于其能够捕捉不同预测领域中复杂的时序依赖关系[9]。这一能力显著提升了网络流量预测的准确性，并使互联网服务提供商能更深入地洞察使用趋势与异常现象。然而，Wu和Keogh[9]指出，由于缺乏长期真实世界数据集，评估预测方法的最新进展仍存在困难。他们表明，依赖合成数据集可能会产生无法推广到实际场景的进步假象。类似地，Ferreira等人也提出了相同观点。[10] 他们指出，缺乏网络流量预测的参考数据集是影响性能评估的主要障碍，并强调大多数方法目前仅在人工生成的数据或非公开可用数据上进行过测试。

The recent release of the CESNET-TimeSeries24 dataset [11] provides real-world ISP network traffic data from thousands of devices at multiple aggregation granularities,

最新发布的 CESNET -TimeSeries24数据集[11]提供了来自数千台设备、涵盖多种聚合粒度的现实世界互联网服务提供商（ISP）网络流量数据。

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hierarchical levels, and with various time series metrics. This dataset enables a level of comprehensive evaluation in network traffic forecasting that was previously impossible with limited or artificially generated datasets. Building on this, our study conducts an in-depth investigation of the performance and limitations of state-of-the-art deep learning models across diverse conditions and settings, while leveraging the unique features of CESNET-TimeSeries24. Using a common, reliable dataset allows for a direct comparison of the strengths and weaknesses of different forecasting models, offering critical insights for the development of forecasting frameworks and systems. Our investigation is guided by the following research questions:

该数据集包含多级层级结构及多种时间序列指标，能够实现网络流量预测领域的全面评估——这在以往使用有限或人工生成的数据集时是无法实现的。在此基础上，本研究深入探讨了前沿深度学习模型在不同场景下的性能与局限性，并充分利用 CESNET -TimeSeries24 的独特功能。通过使用通用且可靠的数据集，可直接比较各类预测模型的优势与不足，为预测框架和系统的开发提供关键见解。本研究围绕以下核心问题展开：

RQ1: What is the performance of state-of-the-art deep learning models for real-world network traffic forecasting?

问题1: 最先进的深度学习模型在实际网络流量预测中的性能如何？

RQ2: How do the training and prediction windows impact the forecasting performance of state-of-the-art deep learning models?

RQ2: 训练窗口与预测窗口如何影响最先进深度学习模型的预测性能？

RQ3: How does the granularity of aggregation (e.g., individual IP addresses, subnets, or institutions) impact the forecasting performance of state-of-the-art deep learning models?

问题3: 聚合粒度（例如单个IP地址、子网或机构）如何影响最先进的深度学习模型的预测性能？

RQ4: How do different monitored metrics (e.g., transmitted data, TCP / UDP ratio, average duration) affect the forecasting performance of state-of-the-art deep learning models?

问题4: 不同监控指标（如传输数据量、TCP/UDP比例、平均持续时间）如何影响先进深度学习模型的预测性能？

The contribution of this study goes beyond comparing advanced forecasting models on a comprehensive dataset. It offers insights that are both model-specific and broadly relevant to the network forecasting domain, enabled for the first time by the availability of a large-scale real-world dataset. In addition, by combining rigorous methodological practices with open-access resources, we establish a fair, transparent, and reproducible benchmark for future research. Our goal is to encourage the adoption of standardized datasets, enabling consistent comparisons and generating practical insights. To further support reproducibility, we release the source code for all models and experiments on GitHub.¹

本研究的贡献不仅在于在综合性数据集上比较了先进的预测模型，更提供了既具有模型特异性又广泛适用于网络预测领域的洞见——这得益于大规模真实世界数据集的首次可用。此外，通过将严谨的方法论与开放获取资源相结合，我们为未来研究建立了一个公平、透明且可复现的基准体系。我们的目标是推动标准化数据集的应用，实现一致性的模型比较并生成实用见解。为进一步保障研究可重复性，我们在GitHub上公开发布了所有模型及实验的源代码。¹

The rest of this paper is structured as follows. Section II describes the related literature. Section III describes the dataset used. Section IV describes selected state-of-the-art deep learning models for forecasting. Section V describes the methodology of the comparison analysis. Section VI describes the results. Section VII discusses the research questions and the lessons learned. Finally, Section VIII summarizes the paper.

本文其余部分结构安排如下：第II节介绍相关文献；第III节阐述所使用的数据集；第IV节介绍用于预测的精选前沿深度学习模型；第V节说明比较分析的方法论；第VI节呈现研究结果；第VII节讨论研究问题及所得启示；最后，第VIII节对全文进行总结。

II. RELATED WORK

II.相关工作

Network traffic forecasting is a critical area of research due to its importance in network optimization, resource allocation, and anomaly detection [2], [3], [10]. Various approaches have been developed over the years, leveraging both statistical and machine learning methods [10].

网络流量预测因其在网络优化、资源分配及异常检测方面的关键作用，已成为一项重要的研究领域[2]。[3], [10]多年来，研究人员结合统计学方法与机器学习方法[10]开发出了多种解决方案。

Currently, statistically-based methods are commonly applied in network traffic forecasting with many published approaches. Such statistically-based methods include Auto-Regressive Integrated Moving Average (ARIMA), Seasonal ARIMA

目前，基于统计的方法已被广泛应用于网络流量预测领域，目前已发表的相关方法众多。这类统计方法包括自回归积分滑动平均模型（ARIMA）和季节性 ARIMA 模型。

(SARIMA), Exponential Smoothing (ETS), or Hidden Markov Models [12], [13], [14], [15]. However, there are several limitations of statistical-based forecasting [10], [16], [17]. Most of them rely on the assumption of linearity, which restricts their ability to model complex, non-linear relationships often present in real-world systems. Methods like ARIMA and regression models require stationarity in data, meaning trends and variance must remain constant over time, necessitating preprocessing that can complicate analysis. Moreover, these models are sensitive to parameter estimation, and their predictive accuracy can degrade in highly dynamic environments where relationships between variables fluctuate. Statistical approaches also struggle to handle multivariate interactions effectively, leading to oversimplified representations of intricate patterns. These limitations collectively underscore the challenges of applying statistical models in scenarios with complex, non-linear, or rapidly changing dynamics.

（SARIMA）、指数平滑法（ETS）或隐马尔可夫模型 [12], [13], [14], [15]然而，基于统计的预测方法存在若干局限性[10]。[16], [17]大多数统计模型都基于线性假设，这限制了它们对现实系统中常见的复杂非线性关系进行建模的能力。诸如 ARIMA 和回归模型等方法要求数据具有平稳性（即趋势和方差随时间保持恒定），因此需要进行预处理，而这往往会使分析过程变得复杂。此外，这些模型对参数估计非常敏感，在变量间关系不断变化的高动态环境中，其预测精度会显著下降。统计方法在处理多变量交互作用方面也存在局限，容易导致对复杂模式的过度简化。这些缺陷共同凸显了在涉及复杂、非线性或快速变化动态特征的情境下应用统计模型所面临的挑战。

Therefore, deep learning has emerged in recent years as a transformative approach to network traffic forecasting. Primarily due to its ability to model non-linear relationships, capture complex dependencies within data, handle non-stationary time series effectively, and provide superior scalability [10], [18].

因此，深度学习近年来已成为网络流量预测领域一种具有变革性的方法。这主要归功于其能够建模非线性关系、捕捉数据中的复杂依赖关系、有效处理非平稳时间序列，并具备卓越的可扩展性[10]。[18]。

Deep learning became increasingly valuable for forecasting, particularly with the introduction of Long Short-Term Memory (LSTM) and Gated Recurrent Units (GRU), which mitigate the exploding and vanishing gradient problems that affected earlier Recurrent Neural Networks (RNN) [19]. These deep learning models have been successfully applied in the network traffic forecasting domain [20], [21]. Nevertheless, the rapid expansion of deep learning forecasting variants in recent years underscores the need for a comprehensive evaluation.

深度学习在预测领域的重要性日益凸显，尤其是随着长短期记忆网络（LSTM）和门控循环单元（GRU）的引入——这些技术有效缓解了早期循环神经网络（RNN）中存在的梯度爆炸与梯度消失问题[19]。此类深度学习模型已成功应用于网络流量预测领域[20]。[21]然而，近年来深度学习预测方法的迅速发展凸显了开展全面评估的必要性。

The first major enhancement of LSTM and GRU models was their integration with Fully Convolutional Networks (FCNs) [22], [23]. Moreover, Temporal Convolutional Networks (TCNs) and their combination with an RNN variant were successfully applied in network traffic forecasting [24]. These hybrid models gained popularity due to their ability to outperform classical approaches while requiring minimal data preprocessing and easily scaling to large time series datasets [25].

LSTM 和GRU模型的首次重大改进在于将其与全卷积网络（FCN）相结合 [22]。[23]此外，时域卷积网络（TCN）及其与 RNN 变体的组合已被成功应用于网络流量预测[24]。这些混合模型因其性能优于传统方法、仅需极少数数据预处理且易于扩展至大型时间序列数据集[25]而广受青睐。

Moreover, more sophisticated deep learning models originally developed for image classification have also been applied in the time series forecasting area. Notable examples include Residual Networks (ResNet) [26]. Building on these architectures, additional models have been developed, such as InceptionTime [27], which integrates residual blocks with Global Average Pooling layers.

此外，最初为图像分类开发的更复杂的深度学习模型也已被应用于时间序列预测领域。典型代表包括残差网络（ResNet）[26]。基于这些架构，研究人员还开发了其他模型，例如InceptionTime[27]——该模型将残差块与全局平均池化层相结合。

Finally, one of the most recent forecasting approaches uses Transformers, which have become a prominent tool in time series forecasting due to their capability to model long-range dependencies, critical for network traffic prediction [28]. However, traditional Transformers face challenges such as inefficiency and a lack of interpretability in capturing temporal dynamics. To address these limitations, several advancements have been proposed. For instance, ETSformer [29] introduces exponential smoothing and frequency attention mechanisms to replace conventional self-attention, enabling the model to capture both seasonal and trend components in a decomposable and interpretable manner. Similarly, models like Informer [30] and Autoformer [31] leverage sparsity and frequency-based approaches to enhance efficiency and adaptability in long-term sequence forecasting. Despite their success, recent studies question the necessity of complex Transformer-based architec-

¹<https://github.com/koumajos/isp-forecasting-benchmark>

tures, with findings that simple linear models can outperform them in several scenarios [32]. Nevertheless, Transformers remain at the forefront of innovations in network traffic forecasting due to their ability to incorporate dynamic time-dependent features and extract meaningful patterns from large-scale, non-stationary data [33], [34].

最后，最近的一种预测方法采用了Transformer模型。该模型因其能够建模长期依赖关系而成为时间序列预测领域的主流工具，这对网络流量预测至关重要[28]。然而，传统Transformer模型在捕捉时序动态方面存在效率低下和可解释性不足等局限。为解决这些问题，学界提出了多项改进方案：例如ETSformer[29]引入指数平滑和频率注意力机制替代传统自注意力机制，使模型能够以可分解且可解释的方式同时捕捉季节性和趋势成分；类似地，Informer[30]和Autoformer[31]等模型则利用稀疏性和基于频率的方法来提升长期序列预测的效率与适应性。尽管这些方法取得了成功，但近期研究对复杂Transformer架构的必要性提出质疑，研究表明在多种场景下简单的线性模型反而能取得更优性能[32]。尽管如此，Transformer模型仍处于网络流量预测创新领域的前沿地位，这得益于其能够整合动态时变特征，并从大规模非平稳数据中提取具有意义的模式[33]。[34]。

The forecasting is usually done separately when multiple time series of multiple processes are present [35]. However, a recent study [36] also studied forecasting of groups of related time series, exploring how algorithms can leverage shared structure among series.

当存在多个不同过程的多个时间序列时，预测通常需要分别进行[35]。然而，近期一项研究[36]也探讨了相关时间序列组的预测问题，分析了算法如何利用序列间的共同结构。

Many of the published works utilize different datasets, which makes approaches hardly comparable. For example, many datasets used in the network traffic forecasting domain were not created with the intention of forecasting evaluation, making them unsuitable for such purposes. An example of this is the WIDE project data in the MAWI archive [37], which contains only 15 minutes per day. Moreover, some related work are built on datasets that are not publicly available [38], [39], making it impossible to independently verify the results. Others rely on synthetic data [40], [41], which, while useful for controlled experiments, fail to provide insights into real-world network environments. Furthermore, even widely used datasets, such as KDD 99, NSL-KDD, Internet2 [42], GEANT [43], and Italia Telecom [44], may lack the comprehensive representation of modern, dynamic traffic patterns required for robust evaluation.

许多已发表的研究均使用不同的数据集，这导致不同研究方法难以直接比较。例如，网络流量预测领域使用的多数数据集并非专为预测评估而设计，因此并不适用于此类用途。MAWI 档案库中的WIDE项目数据[37]便是典型例证——其每日数据量仅15分钟。此外，部分相关研究所使用的数据集也未公开共享[38]。[39]这使得无法独立验证实验结果。其他研究则依赖合成数据[40]。[41]这类数据集虽适用于受控实验，却无法深入揭示真实网络环境的特性。此外，即便是广泛使用的数据集（如 KDD 99、NSL-KDD 、Internet2[42]、 GEANT[43] 及 Italia Telecom [44]），也可能缺乏对现代动态流量模式的全面描述——而这正是进行可靠评估所必需的条件。

The recent release of a large, standardized, and publicly available real-world dataset, CESNET-TimeSeries24 [11], provides a valuable resource for comparing existing forecasting models. This comprehensive dataset enables a more consistent evaluation of deep learning approaches in realistic network scenarios. By leveraging this dataset, our study aims to perform a comparative analysis of state-of-the-art deep learning models, filling a crucial gap in the domain by establishing more reliable and meaningful performance assessment.

近期发布的大型标准化公开现实数据集 CESNET -TimeSeries24[11]为比较现有预测模型提供了宝贵资源。该全面的数据集能够更一致地评估深度学习方法在真实网络场景中的表现。基于此数据集，本研究旨在对最先进的深度学习模型进行对比分析，通过建立更可靠且具有实际意义的性能评估体系，填补该领域的重要空白。

III. DATASET

三.数据集

To evaluate state-of-the-art deep learning models for real-world network traffic forecasting, we selected a dataset from an actual operational network. The CESNET-TimeSeries24 dataset [11] captures network traffic data from the ISP network CESNET3.²

为评估最先进的深度学习模型在实际网络流量预测中的应用效果，我们从一个实际运营的网络中选取了数据集。CESNET -TimeSeries24数据集[11]收录了来自互联网服务提供商CESNET3的网络流量数据²。

The dataset is uniquely comprehensive, offering multivariate time series created through traffic aggregation at three distinct intervals: 10 minutes, 1 hour, and 1 day. Each interval-specific time series contains multiple metrics useful for network traffic forecasting and anomaly detection. Given that the original capture interval was 10 minutes, metrics that represent unique

该数据集具有独特的全面性，提供了通过三种不同时间间隔（10分钟、1小时和1天）对交通流量进行聚合后生成的多变量时间序列。每个特定时间间隔的时间序列均包含多种可用于网络流量预测和异常检测的指标；鉴于原始采集间隔为10分钟，这些指标能够反映该时间段内的独特特征。

²The Czech Educational and Science Network.

²捷克教育与科学网络。

TABLE I

表 I
AVAILABLE TIME SERIES METRICS FOR 1 HOUR AGGREGATION IN THE CESNET-TIMESERIES24 DATASET
CESNET-TIMESERIES24数据集中用于 1HOUR 聚合的可用时间序列指标

Time series metric name 时间序列指标名称	Description 描述
n_flows	Number of IP flows IP流量数量
n_packets	Number of packets 数据包数量
n_bytes	Number of bytes 字节数
sum_n_dest_ip	Sum of number of unique destination IPs 唯一目标IP地址数量之和
avg_n_dest_ip	Avg. of number of unique destination IPs 唯一目标IP地址数量的平均值
std_n_dest_ip	Std of number of unique destination IPs 唯一目标IP地址数量的标准差
sum_n_dest_ports	Sum of number of unique destination ports 唯一目标端口数量之和
avg_n_dest_ports	Avg. of number of unique destination ports 唯一目标端口数量的平均值
std_n_dest_ports	Std of number of unique destination ports 唯一目标端口数量的标准差
sum_n_dest_asn	Sum of number of unique destination ASNs 唯一目标ASN数量之和
avg_n_dest_asn	Avg. of number of unique destination ASNs 唯一目标ASN数量的平均值
std_n_dest_asn	Std of number of unique destination ASNs 唯一目标ASN数量的标准差
tcp_udp_ratio_packets	TCP/UDP packet ratio TCP/UDP数据包的比例
tcp_udp_ratio_bytes	TCP/UDP bytes ratio TCP/UDP字节比率
dir_ratio_packets	Directional ratio of packets 数据包的方向性比率
dir_ratio_bytes	Directional ratio of bytes 字节的方向性比率
avg_duration	Average duration 平均时长
avg_ttl	Average TTL 平均TTL

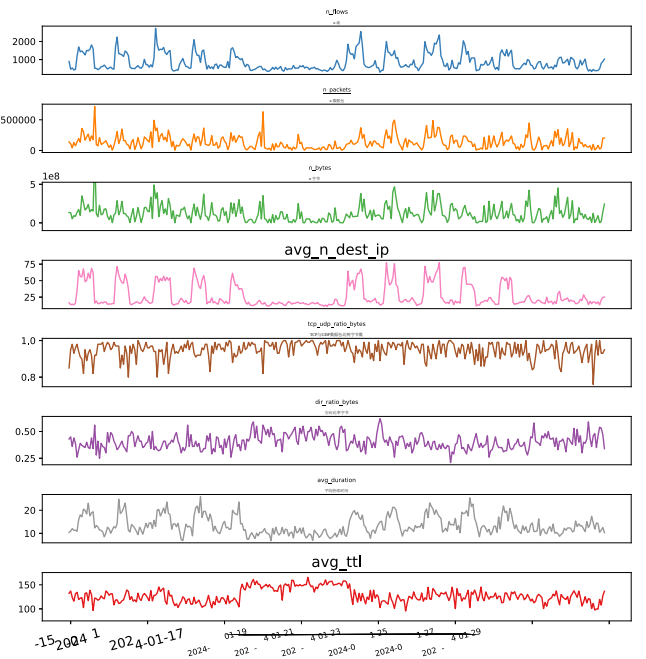


Fig. 1. Example of time series metrics aggregated by 1 hour from the CESNET-TimeSeries24 dataset.

图1.来自 CESNET -TimeSeries24数据集、按每小时聚合的时间序列指标示例。

values were adapted for the 1 hour and 1 day series incorporating sum, average and standard deviation statistics to enhance temporal granularity. We limit this study to 1 hour aggregation interval. The time series metrics for 1 hour aggregation interval are listed in Table I. The example multivariate time series aggregated by 1 hour is shown in Figure 1.

我们针对1小时和1天时间序列对数值进行了调整，采用总和、均值及标准差等统计量以提升时间粒度。本研究仅限于1小时聚合区间。1小时聚合区间的时序指标列于表I中；按1小时聚合的多变量时序示例见图1。

A standout feature of the CESNET-TimeSeries24 dataset is its extensive coverage and granularity. The dataset encompasses not only general network traffic but also highly detailed time series across 283 institutions, 548 institutional subnets, and over 270,000 individual IP addresses. This extensive range provides a robust basis for comparative analysis of deep learning models, enabling researchers to benchmark forecasting performance across multiple hierarchical levels within the network. This dataset enables a detailed examination of the scalability, adaptability, and performance of models in forecasting diverse traffic patterns within a real-world ISP environment. Spanning a substantial 40-week period, from October 2023 to July 2024, the dataset captures both long-term trends and fine-grained fluctuations, making it suitable for comprehensive performance evaluation. The examples of different behaviors of time series of institutions, institution

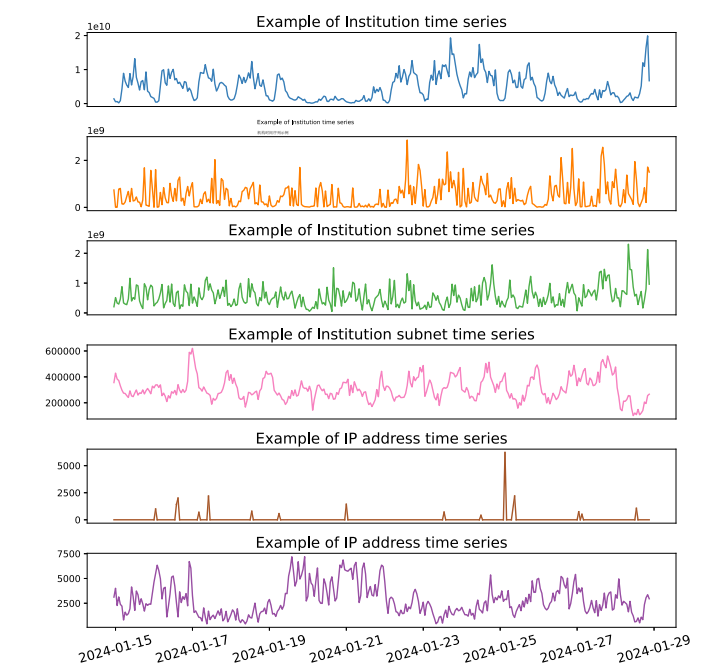


Fig. 2. Example of different hierarchical levels in time series metric *Number of bytes* aggregated by 1 hour from the CESNET-TimeSeries24 dataset.

图2. 时间序列指标中不同层级结构的示例CESNET-TimeSeries24数据集中按每小时聚合的字节数。

subnets, and IP addresses are shown in Figure 2.

CESNET-TimeSeries24数据集的一个显著特点是其广泛的覆盖范围和精细的粒度。该数据集不仅包含通用网络流量数据，还涵盖了来自283个机构、548个机构子网以及超过27万个独立IP地址的高精度时间序列数据。这种广泛的数据覆盖为深度学习模型的比较分析提供了坚实基础，使研究人员能够对比评估网络中不同层级结构下的预测性能。该数据集可深入分析模型在真实互联网服务提供商（ISP）环境中预测各类流量模式时的可扩展性、适应性及性能表现。数据集时间跨度长达40周（从2023年10月至2024年7月），既捕捉了长期趋势，也记录了细微波动，非常适合进行全面性能评估。图2展示了各机构、机构子网及IP地址时间序列数据的不同特征示例。

IV. DEEP LEARNING MODELS FOR FORECASTING

IV. 用于预测的深度学习模型

This section provides a brief description of model types or particular state-of-the-art deep learning model architectures chosen to be compared in this study. The types were chosen to represent versatility in time series prediction handling tasks and to vary in computational demands.

本节简要介绍了本研究中选定用于比较的模型类型或特定前沿深度学习模型架构。这些类型的选择旨在体现其在时间序列预测任务处理方面的多样性，并满足不同的计算需求。

Moreover, we provide information about their general advantages and shortcomings that can be found in related work. These models were selected due to their distinct architectural approaches, allowing for a comparative analysis of their strengths and limitations in different predictive contexts. Given the diverse requirements of real-time network operations, the models range from resource-intensive deep learning architectures to more streamlined alternatives, each offering unique benefits and trade-offs. Understanding these differences is essential to assessing the deployability of models in real-world network environments, where both computational efficiency and prediction accuracy are critical factors.

此外，我们还提供了相关研究中已阐述的这些模型的整体优势与不足之处。选择这些模型是基于其独特的架构设计，从而能够对它们在不同预测场景下的优缺点进行对比分析。鉴于实时网络运营环境的需求多样性，这些模型涵盖了从资源密集型深度学习架构到更为精简的替代方案等多种类型，每种模型都具有独特的优缺点。理解这些差异对于评估模型在实际网络环境中的适用性至关重要——在现实应用中，计算效率与预测准确性都是关键考量因素。

Mean model used as a baseline for evaluating deep learning models, computes the average value of datapoints from the training window, and applies this mean as the prediction for each datapoint in the prediction window. While its simplicity is advantageous for benchmarking, it can capture only trends and fails to capture temporal patterns and seasonality, making it unsuitable for datasets with significant time dependencies. Nevertheless, its role as a baseline is crucial, as it sets a minimal performance threshold that deep learning models should surpass, ensuring their predictive capacity

is not inferior to a simplistic, non-adaptive approach. **SARIMA** (Seasonal AutoRegressive Integrated Moving Average) is a statistical model for time series forecasting. SARIMA is interpretable and effective for data with strong linear trends and seasonal structures. However, due to reliance on linear assumptions, SARIMA cannot capture complex nonlinear relationships in data. SARIMA is used to compare deep learning models with a classical statistical approach.

均值模型被用作评估深度学习模型的基线方法，它计算训练窗口内数据点的平均值，并将该均值作为预测窗口中每个数据点的预测结果。虽然其简洁性在基准测试中具有优势，但仅能捕捉趋势而无法反映时间序列中的模式和季节性特征，因此不适用于存在显著时间依赖性的数据集。不过，其作为基线模型的作用至关重要，因为它设定了深度学习模型必须超越的最低性能阈值，确保其预测能力不低于简单的非自适应方法。 **SARIMA**（季节性自回归积分滑动平均）是一种用于时间序列预测的统计模型。 SARIMA 对具有明显线性趋势和季节结构的数据具有可解释性且效果显著；但由于依赖线性假设， SARIMA 无法捕捉数据中的复杂非线性关系。 SARIMA 常被用于将深度学习模型与经典统计方法进行比较。

LSTM (Long Short-Term Memory) networks are well-regarded for their capacity to capture long-term dependencies in sequence data, as they mitigate the vanishing gradient problem that hinders traditional RNNs. The architecture utilizes memory cells, which allow the model to retain information over extended time steps, making it highly effective for time-series prediction tasks. However, LSTMs are computationally intensive and require substantial memory resources, which can be a limitation in environments with constrained computational power. Additionally, they tend to have a large parameter count, which can lead to overfitting, especially when working with smaller datasets.

长短期记忆（LSTM）网络因其能够捕捉序列数据中的长期依赖关系而备受推崇，它有效缓解了传统循环神经网络面临的梯度消失问题。该架构采用记忆单元设计，使模型能够在较长的时间步长内保留信息，因此在时间序列预测任务中表现尤为出色。然而，LSTM具有较高的计算复杂度且需要大量内存资源，在计算能力受限的环境中可能成为制约因素；此外其参数量通常较大，尤其在处理小型数据集时易导致过拟合现象。

GRU (Gated Recurrent Unit) networks are a simplified alternative to LSTM networks, designed to efficiently capture dependencies in sequential data while addressing the vanishing gradient problem. GRUs streamline the architecture by combining the forget and input gates into a single update gate, which reduces the parameter count and computational demands. This makes GRUs faster to train and more memory-efficient, especially on smaller datasets or resource-constrained environments. However, their simplified structure may limit their ability to model complex long-term dependencies as effectively as LSTMs, which can be a drawback for tasks requiring extended temporal context.

GRU（门控循环单元）网络是 LSTM 网络的一种简化替代方案，旨在高效捕捉序列数据中的依赖关系，并解决梯度消失问题。GRU通过将遗忘门和输入门整合为单一更新门来优化架构，从而减少参数数量并降低计算需求。这使得GRU训练速度更快且更节省内存，尤其适用于小型数据集或资源受限的环境。然而，其简化的结构可能限制其像LSTM那样有效建模复杂长期依赖关系的能力，这对需要处理长时序上下文的任务而言可能存在不足。

LSTM-FCN model [22] combines the strengths of LSTM networks with Fully Convolutional Networks (FCNs), enabling the model to capture both temporal and spatial features in time-series data. This hybrid architecture improves feature extraction and helps generalize better to complex data patterns. Nevertheless, the increased complexity of combining LSTM with FCN layers results in greater computational demands and can make training slower, especially on larger datasets or limited hardware.

LSTM-FCN模型[22]融合了 LSTM 网络与全卷积网络（FCN）的优势，使该模型能够同时捕捉时间序列数据中的时序特征与空间特征。这种混合架构提升了特征提取能力，并有助于更好地泛化复杂数据模式。然而，将 LSTM 层与 FCN 层结合所带来的复杂性增加会导致更高的计算需求，尤其在处理大规模数据集或使用有限硬件资源时，训练速度可能会显著下降。

GRU-FCN model [23] integrates GRUs with Fully Convolutional Networks, blending the benefits of a simpler recurrent architecture with powerful feature extraction capabilities. GRUs offer a streamlined alternative to LSTMs, potentially reducing training time due to their more efficient structure. Nevertheless, GRU-FCN may not capture long-term dependencies with the same depth as LSTM models, which can **GRU-FCN**模型[23]将GRU与全卷积网络相结合，既保留了简单循环架构的优势，又具备强大的特征提取能力。相较于LSTM，GRU是一种更为简洁的替代方案，其高效结构有望缩短训练时间。然而，GRU-FCN可能无法像 LSTM 模型那样准确捕捉长期依赖关系。

be a drawback when handling complex, long-sequence data.

在处理复杂、长序列数据时会成为缺点。

ResNet models [26] are known for their residual connections, which help mitigate the vanishing gradient problem in deep neural networks, making them well-suited for extracting hierarchical features in time-series data. The residual connections enable training deeper architectures without the degradation issues typical in very deep networks. Nonetheless, ResNets can be resource-intensive and may require a significant amount of data to reach their full potential, as they are prone to overfitting on smaller datasets.

ResNet模型[26]以其残差连接结构而闻名，这种结构有助于缓解深度神经网络中的梯度消失问题，使其非常适合从时间序列数据中提取层次化特征。残差连接使得训练更深层的网络架构时不会出现超深网络常见的性能下降问题。然而，ResNet模型对计算资源要求较高，并且需要大量数据才能充分发挥其潜力，因为它们容易在较小数据集上发生过拟合现象。

InceptionTime model [27] is a deep convolutional model inspired by the Inception architecture, specifically adapted for time-series classification. Its multi-scale feature extraction capabilities allow it to capture both local and global patterns in time-series data, contributing to its high performance across various benchmark tasks. However, the model’s depth and complexity make it computationally expensive and may require extensive tuning to achieve optimal results, especially in cases with less structured or lower-dimensional data.

InceptionTime模型[27]是一种受 Inception 架构启发的深度卷积模型，专门针对时间序列分类任务进行优化。其多尺度特征提取能力能够同时捕捉时间序列数据中的局部与全局模式，从而使其在各类基准测试中表现优异。然而，该模型的复杂结构和深度要求导致计算成本较高，尤其在处理结构化程度较低或低维数据时，可能需要进行大量参数调优才能获得最佳效果。

RCLSTM (Random Connectivity LSTM) model [45] is an optimized variant of LSTM designed to reduce computational complexity while maintaining comparable predictive performance. RCLSTM introduces stochastic connectivity to conventional LSTM neurons, creating a sparse neural structure that significantly decreases the number of trainable parameters. This modification improves computational efficiency, making the model particularly suitable for latency-stringent and power-constrained environments. Despite the reduction in connections, RCLSTM preserves long-range dependencies effectively, making it highly applicable in tasks such as traffic forecasting and user mobility prediction.

RCLSTM（随机连接 LSTM）模型[45]是 LSTM 的一种优化版本，旨在降低计算复杂度的同时保持相当的预测性能。RCLSTM 在传统 LSTM 神经元中引入随机连接机制，形成稀疏神经结构，显著减少可训练参数数量。这一改进提升了计算效率，使该模型特别适用于对延迟要求严格且功耗受限的场景。尽管连接数量减少，RCLSTM 仍能有效保留长距离依赖关系，因此非常适用于交通预测和用户移动性预测等任务。

ETSformer model [29] is a deep learning model that integrates principles of Exponential Smoothing (ETS) into the Transformer architecture. By embedding trend and seasonal decomposition into the Transformer framework, ETSformer enhances interpretability while retaining the strengths of self-attention. This design allows it to capture both short-term fluctuations and long-term patterns more effectively. However, ETSformer’s architecture is more complex than traditional methods, and training can be computationally demanding compared to other approaches.

ETSformer模型[29]是一种将指数平滑（ETS）原理融入Transformer架构的深度学习模型。通过将趋势分解和季节性分解整合到Transformer框架中，ETSformer在保持自注意力机制优势的同时提升了可解释性，从而更有效地捕捉短期波动与长期趋势。然而，其架构相较于传统方法更为复杂，训练过程对计算资源的要求也高于其他方法。

The models and architectures discussed in this section are commonly used in the time series prediction domain, each providing promising robustness and modelling potential for this task, while varying in computational complexity. The variability in the chosen models should demonstrate the potential for their application in real-world network operations, where one has to balance between accuracy and computational efficiency.

本节讨论的各类模型与架构均广泛应用于时间序列预测领域，每种模型均具备出色的鲁棒性和建模能力，同时计算复杂度各不相同。所选模型的多样性充分体现了其在实际网络运营中的应用潜力——在实际应用中，必须在预测精度与计算效率之间取得平衡。

V. METHODOLOGY
V. 研究方法

Our methodology follows the recommended usage of the CESNET-TimeSeries24 dataset [11]. We use multiple dataset parts provided in the dataset—institutions, institution subnets, and an IP address sample. We trained models on time series aggregated on 1 hour time intervals, and each time series metric was modelled separately. Thus, we provide a comparison



Fig. 3. Holistic view of methodology applied to each time series metric.
图 3. 应用于各时间序列指标的方法论整体示意图。

of only univariate approaches. We evaluate models on all time series metrics available in the dataset.

我们的方法遵循 CESNET -TimeSeries24数据集的推荐使用规范[11]。我们采用了该数据集中提供的多个数据集部分——机构信息、机构子网络以及IP地址样本。模型基于每小时时间间隔聚合的时间序列进行训练，且每个时间序列指标均独立建模。因此，我们仅提供单变量分析方法的对比结果，并在数据集中所有可用的时间序列指标上对模型进行评估。

Pre-processing involves several steps for each time series, as shown in Figure 3, including data splitting, standardizing the data, and addressing missing values.

如图3所示，预处理针对每个时间序列均包含多个步骤，包括数据分割、数据标准化以及缺失值处理。

A. Missing Values Handling

A. 缺失值的处理

Real-world network traffic is usually not constantly generated, especially in the case of monitoring a single device. Thus, it is inevitable that the collected time series contains missing values [46], which was identified as one of the challenges of the dataset [11].

实际网络流量通常并非持续生成，尤其是在监测单个设备时。因此，采集到的时间序列数据必然存在缺失值[46]，这已被视为该数据集面临的主要挑战之一[11]。

In this comparative study, we fill missing values with zeros for most metrics. Since missing values in the dataset represent an interval where no data was transmitted. Thus, the zero value represents true information for metrics such as the number of IP flows, packets, and bytes. The zero values will force the model to learn how to predict when the device is not communicating. However, modeling time series that naturally contain zero values is a non-trivial challenge [47], [48]. Therefore, one of the objectives of this study is to verify how DL models handle time series with filled missing values. Because in the dataset, there are extreme cases where zero values represent a majority of values. The direction and TCP/UDP ratios are filled with 0.5 values, as they represent a true balance where no data is transmitted.

在这项比较研究中，我们为大多数指标的缺失值均填入零值。由于数据集中的缺失值代表未传输数据的时间区间，因此对于IP流量、数据包及字节数等指标而言，零值即代表真实信息。这些零值将促使模型学习如何预测设备处于非通信状态。然而，对天然包含零值的时间序列进行建模本身就是一个非同小可的挑战[47]。[48]因此，本研究的目标之一是验证深度学习模型如何处理存在填充缺失值的时间序列数据。由于该数据集中存在零值占绝大多数的极端情况，方向参数及TCP/UDP比例均被填充为0.5值，以反映实际无数据传输时的真实平衡状态。

B. Time Series Splitting, Normalization, and Sliding Window

B. 时间序列分割、标准化与滑动窗口处理

Each time series is split into train, validation, and test sets for the correct model training and evaluation. The duration of the training set represents the first 35% of the time series (14 weeks). The validation set holds the next 5% (2 weeks), and the remainder (24 weeks) is used as the test set.

每个时间序列均被划分为训练集、验证集和测试集，以确保模型训练与评估的准确性。训练集涵盖时间序列的前35%（14周）；验证集占后续5%（2周）；剩余部分（24周）则作为测试集。

At first, we fill in missing time series values with zeros. Then, we train the MinMax scaler on the train set and transform the train, validation, and test parts of the data using the fitted scaler.

首先，我们用零值填补缺失的时间序列数据。随后，在训练集上训练MinMax缩放器，并利用该拟合后的缩放器对数据的训练集、验证集和测试集进行转换。

We used the sliding-window approach, which is a widely adopted method for splitting data into multiple training samples from a single continuous time series stream. In our methodology, each neural network has an input of size equal to the number of consecutive time series values in a training window (also often referred to as a look-back window). The training window has a fixed length from which the model learns temporal dependencies. Based on this input, the model generates a specified number of consecutive future values. The number of predicted future values depends on a predefined prediction window size (also often referred to as a forecast horizon).

我们采用了滑动窗口方法，这是一种广泛应用于从单一连续时间序列数据流中划分多个训练样本的经典方法。在我们的方法论中，每个神经网络的输入长度等于训练窗口（通常称为回溯窗口）内连续时间序列值的数量。该训练

TABLE II
表 II
USED TRAINING AND PREDICTION WINDOWS
使用的训练窗口与预测窗口

Training window size	Prediction window size
24	1
168	1
168	24
744	1
744	168

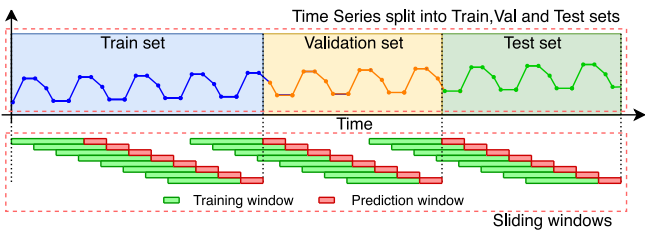


Fig. 4. Illustration of time series partitioning into training, validation, and test sets, along with the sliding window approach for generating overlapping training and prediction windows.
图4. 时间序列划分为训练集、验证集和测试集的示意图，以及用于生成重叠训练窗口与预测窗口的滑动窗口方法。

窗口具有固定长度，模型正是通过分析这些数据来学习时间依赖关系。基于此输入数据，模型可生成指定数量的连续未来预测值。预测值的数量取决于预设的预测窗口大小（通常称为预测范围）。
The specific combinations of the training window size and the prediction window size evaluated in this study are summarized in Table II. We have chosen training and prediction window sizes commonly used in related work [10], [49]. In this comparative study, we systematically adjust the training and prediction window sizes to the day, week, and month. Thus, we explore how the performance of models varies depending on the amount of historical data considered and the length of the forecast.

本研究评估的训练窗口大小与预测窗口大小的具体组合如表II所示。我们采用了相关文献[10]中常用的训练窗口和预测窗口尺寸。[49]在这项比较研究中，我们系统地根据日期、周数和月份数调整训练窗口与预测窗口的大小。由此，我们探究了模型性能如何随所使用的历史数据量及预测周期长度的不同而变化。

Once a training–prediction pair is formed, the window is shifted forward by a prediction window. This process is repeated across the entire dataset, creating overlapping sequences that capture both short-term and long-term variations in the data. As a result, the model is trained on diverse input-output examples rather than relying on a single continuous segment. Thereby, it improves the ability of models to generalize.

一旦形成训练样本与预测结果的配对数据，模型会将预测窗口向前移动一个预测步长。这一过程会在整个数据集中重复执行，从而生成相互重叠的序列，全面捕捉数据中的短期及长期变化特征。这样一来，模型便能基于多样化的输入-输出样本进行训练，而非仅依赖单一连续数据片段；这显著提升了模型的泛化能力。

In this study, the sliding window approach is applied consistently to the training, validation, and test sets of the time series, as illustrated in Figure 4.

本研究中，滑动窗口方法被一致应用于时间序列的训练集、验证集和测试集，如图4所示。

C. Models Optimization
C. 模型优化

In this work, we are finding the optimal model architecture, which can perform best in general when considering all time series. Because varying hyperparameters specifically tuned for each separate time series is often inapplicable in real-world scenarios [50].

在本研究中，我们旨在寻找一种最优的模型架构，该架构在处理所有时间序列时均能取得最佳性能。因为在现实场景中，针对每个独立时间序列分别调整不同的超参数通常是不可行的[50]。

During the training of DL models, the Adam optimizer was employed. We choose the Adam due to its adaptive learning rate mechanism and momentum-based updates, which generally improve convergence speed and training stability compared to standard stochastic gradient descent [51].

在深度学习模型的训练过程中，我们采用了Adam优化器。选择Adam优化器是基于其自适应学习率机制及基于动量的更新策略；与标准随机梯度下降方法相比，这些特性通常能显著提升收敛速度和训练稳定性[51]。

We use MSELoss as the loss function to measure discrepancies between predicted and actual values. A hyperparameter search was conducted to explore various architectural and optimization configurations. This search includes the number

TABLE III
表 III
SUMMARY OF FINAL HYPERPARAMETERS
最终 HYPERPARAMETERS 摘要

Model	RNN layers	Hidden size	Batch	Learning rate	Epochs
GRU	1	100	16	0.01	100
LSTM	1	100	16	0.01	100
GRU-FCN	4	20	16	0.01	20
LSTM-FCN	1	100	16	0.01	100
InceptionTime	-	-	128	0.001	20
ResNet	-	-	128	0.01	20
RCLSTM	1	300	32	0.01	100

of RNN layers, hidden layer size, batch size, learning rate, and number of epochs. The tested number of RNN layers and hidden size were chosen for each model based on related work that uses the model for network traffic prediction. Most of the models were trained with 1 to 5 RNN layers and hidden layer sizes from 20 to 200. We perform a search for optimal architecture on this scale of the number and sizes of layers. The final architecture and hyperparameter settings are presented in Table III.

我们采用MSELoss作为损失函数来衡量预测值与实际值之间的差异。通过超参数搜索探索了多种架构和优化配置方案，搜索参数包括 RNN 层数量、隐藏层大小、批量大小、学习率以及训练轮数。针对每个模型所选用的 RNN 层数量及隐藏层大小均参考了该模型在网络流量预测领域的相关研究结果。大多数模型均使用1至5个 RNN 层，并采用20至200不等的隐藏层大小进行训练。我们在这一层数与层大小范围内进行了最优架构搜索，最终确定的架构方案及超参数设置详见表III。

These hyperparameters were selected based on the average results on a random sample of 50 time series from each dataset part. This setup of hyperparameter search allows finding a model setting that works generally well for network traffic prediction, regardless of the hierarchical level of aggregation or time series metric. Most of the models ended with one RNN layer, which may be surprising at first glance. However, it is in line with the findings of Zeng et al. [32]. In their work, they prove that a simple model can outperform even complicated models such as a Transformer-based model. For the LSTM-FCN and GRU-FCN models, we also tuned the sizes of CNN layers. Both model architectures contain three CNN layers with resulting best-performing sizes 128, 256, and 128 for LSTM-FCN, and 64-128-64 for GRU-FCN.

这些超参数是根据从每个数据集部分随机抽取的50个时间序列样本的平均结果确定的。这种超参数搜索方法能够找到一种普遍适用于网络流量预测的模型配置，无论聚合层级或时间序列度量如何。大多数模型均采用单层 RNN 结构，乍看可能令人意外，但这与Zeng等人的研究结论一致。[32]在他们的研究中，他们证明了一个简单的模型甚至能够超越基于Transformer的复杂模型。对于 LSTM - FCN 和GRU- FCN 模型，我们还调整了CNN层的数量：两种模型架构均包含三个CNN层，其中 LSTM - FCN 的最佳层数为128、256和128层；GRU- FCN 则为64-128-64层。

For the RCLSTM model, the architecture and hyperparameters were taken from the original publication, as their methodology also performs hourly forecasting. Likewise, InceptionTime, ResNet, and ETSformer were deployed with their standard, predefined architectures. By consolidating and refining the results of multiple experimental trials, the chosen hyperparameters perform best as evaluated across a range of time series in used datasets parts, as presented in the next section.

对于 RCLSTM 模型，其架构和超参数均取自原始论文，因为该方法同样支持每小时预测。同样地，InceptionTime、ResNet和ETSformer均采用其标准预定义架构进行部署。通过整合并优化多次实验结果，所选超参数在所用数据集的各类时间序列数据上均表现出最佳性能，具体如下一节所示。

For the SARIMA model, we train parameters using the same methodology with all terms ranging from 1 to 5 and season ranging from one datapoint to one week of datapoints. The best performing model on the validation set of time series was SARIMA(1,1,1)(1,1,1,1).

对于 SARIMA 模型，我们采用相同方法训练参数，所有参数项取值范围为1至5，季节维度则从单个数据点到一周数据点不等。在时间序列验证集上表现最佳的模型为 SARIMA(1,1,1)(1,1,1,1)。

D. Evaluation Measures
D. 评估方法

The evaluation follows a common procedure where we forecast the next values based on previous ones using the trained models. To evaluate the precision of the forecasting, we use two common fully unsupervised measures, Root Mean Square Error (RMSE) and R^2 -score (also called the coefficient of determination). In the following equations, y represents the actual normalized value, while $\hat{y}$ represents the predicted value. The RMSE is defined as:

该评估遵循一种通用流程：我们利用训练好的模型，根据先前数据预测后续值。为评估预测精度，我们采用两种常见的完全无监督评价指标——均方根误差（RMSE）和 R^2 分数（亦称决定系数）。在以下公式中， y 表示实际归一化值， $\hat{y}$ 表示预测值。RMSE 的定义如下：

$$\text{RMSE}(y, \hat{y}) = \sqrt{\frac{1}{N} \sum_{i=0}^{N-1} (y_i - \hat{y}_i)^2}. \quad (1)$$

The RMSE ranges from 0 to ∞ , with lower values indicating more accurate predictions. The measure highlights models that do not have large errors in prediction. However, its sensitivity to large errors can be insufficient, sometimes obscuring the distinction between consistent underperformance and occasional extreme errors.

RMSE 的取值范围从 0 到 ∞ ，数值越低表示预测精度越高。该指标能凸显预测误差较小的模型；然而其对较大误差的敏感度可能不足，有时难以区分持续性表现不佳与偶发性极端误差。

The R^2 -score measure (also called the coefficient of determination) is a statistical measure that shows the proportion of the dependent variable's variance explained by the independent variable, calculated using the following equation:

R^2 分数指标（亦称为决定系数）是一种统计量，用于表示自变量对因变量方差的解释比例，其计算公式如下：

$$R^2(y, \hat{y}) = 1 - \frac{\sum_{i=1}^n (y_i - \hat{y}_i)^2}{\sum_{i=1}^n (y_i - \mu_y)^2 + \epsilon}. \quad (2)$$

The R^2 -score ranges from $-\infty$ to 1, with higher values indicating more accurate predictions. The R^2 -score measure estimates goodness-of-fit, showing how well the model captures overall data trends.

R^2 分数的取值范围为 $-\infty$ 至 1，数值越高表示预测越准确。 R^2 分数用于评估模型的拟合优度，反映模型对整体数据趋势的捕捉能力。

Related work usually reports both RMSE and R^2 -score because of their distinguishing properties [52]. However, combining RMSE and R^2 -score measures offers a comprehensive evaluation of forecasting models by capturing different aspects of performance. Some studies combine various forecasting measures, for example, by harmonic mean [53]. RMSE emphasizes large individual errors and highlights variance, while the R^2 -score reflects how well the model explains the variance in the dependent variable.

相关研究通常同时报告 RMSE 和 R^2 分数，因为它们具有独特的区分特性[52]。然而，结合 RMSE 与 R^2 分数指标能够通过涵盖模型性能的不同维度，对预测模型进行全面评估。部分研究会综合多种预测指标（例如采用调和平均值[53]）。RMSE 侧重于衡量单个误差的大小并突出方差；而 R^2 分数则反映模型对因变量方差的解释能力。

To balance these contributions, we introduce a new combined measure that we call *Harmonic score*. This new measure combines the RMSE, giving insight into prediction errors, and the R^2 -score capturing trends, the two measures that we care about when forecasting network traffic. It uses the harmonic mean to penalize extreme values and ensure both metrics influence the final score similarly. The *Harmonic score* measure is defined as:

为平衡这些贡献因素，我们提出了一种新的综合指标，称为*调和分数*。该指标融合了能够反映预测误差的 RMSE 值与捕捉趋势特征的 R^2 分数——这两项是我们预测网络流量时重点关注的指标；同时采用调和均值来惩罚极端值，确保两项指标对最终得分的影响程度一致。*调和分数* 的具体定义如下：

$$\text{Harmonic score} = 2 \frac{\text{RMSE} |R^2 - 1|}{\text{RMSE} + |R^2 - 1|}. \quad (3)$$

Since RMSE and R^2 -score operate on different scales, we rescale them by clipping RMSE at a maximum of 11 and R^2 -score at a minimum of -10 . These thresholds were chosen based on the distribution of model results. The distribution follows approximately a normal distribution. We choose $\mu + 2\sigma$ to remove outliers. Therefore, this captures the central 97.8% of the data to limit the influence of outliers while preserving typical values. This adjustment allows for a meaningful combination using the harmonic mean without distorting the overall evaluation.

由于 RMSE 和 R^2 -得分作用于不同的量值范围，我们通过将 RMSE 截断至最大值 11、将 R^2 -得分截断至最小值 -10 来进行重新缩放。这些阈值的选择基于模型结果的分布情况：该分布近似服从正态分布。我们采用 $\mu + 2\sigma$ 作为剔除异常值的基准值，从而覆盖数据集中 97.8% 的中心部分，既限制异常值的影响又保留典型数值。这种调整使得在使用调和平均数进行综合评估时既能保持数据意义，又不会扭曲整体评价结果。

This equation adjusts the R^2 -score by transforming it to align with RMSE, ensuring both measures are compatible. The harmonic mean emphasizes instances where both measures indicate precision, offering a clear, balanced evaluation.

该方程通过将 R^2 分数转换为与 RMSE 一致的形式，从而调整其数值，确保两种度量标准相互兼容。调和平均值着重反映两种度量均显示高精度的情况，提供清晰且平衡的评估结果。

VI. EXPERIMENTAL FORECASTING RESULTS

六. 实验预测结果

We evaluate the selected DL models across each time series metric and hierarchy level, leading to a large volume of results. However, due to space limitations, not all results can be presented in this paper. Therefore, we focus on presenting the results of the metric *Number of bytes* (transmitted data), as it is commonly used in the network traffic forecasting domain [10]. The results for the remaining features are presented briefly, focusing on key insights. All results and analyses are available in the supplementary materials on the GitHub repository.³

我们针对每个时间序列指标和层次层级对所选深度学习模型进行了评估，从而获得了大量结果。然而由于篇幅限制，本文无法呈现所有结果。因此，我们重点展示字节数（传输数据量）这一指标的结果，因其在网络流量预测领域被广泛使用[10]。其余特征的分析结果均以简要形式呈现，着重突出关键发现。所有结果与分析均可在 GitHub 仓库的补充材料中获取。³

We trained 10 models (Mean, SARIMA, and 8 DL models) across five combinations of training and prediction windows, for 18 time series metrics within 1,831 time series (283 time series for institutions, 548 time series for institutional subnets, and 1000 time series for IP addresses), resulting in 1,647,900 final models. In this section, we present the results for the *Number of bytes* time series metrics (transmitted data).

我们针对 1,831 个时间序列（包括 283 个机构时间序列、548 个机构子网时间序列以及 1000 个 IP 地址时间序列），在五种不同的训练窗口与预测窗口组合下，训练了 10 个模型（均值模型、SARIMA 模型及 8 个深度学习模型），共获得 1,647,900 个最终模型。本节重点展示字节数时间序列指标（传输数据）的分析结果。

A. Detailed Models Performance on Metric Number of Bytes

Tables IV and V present aggregated results for RMSE and R^2 -score respectively. Each row represents the mean of one experimental setting with a specific training and prediction window combination. The result is aggregated over all time series within the dataset time series type (e.g., Institutions, Institutions subnets, and single IPs time series). The best-performing model is highlighted with a bold font and green-shaded cell, while the worst-performing model is marked with a red-shaded cell. Lighter shades indicate second

4. 基于字节数指标的模型性能详细对比表 IV 和 V 分别展示了 RMSE 和 R^2 -score 的汇总结果。每行数据代表采用特定训练窗口与预测窗口组合时单次实验设置的均值，结果汇总自数据集中所有时间序列类型（如机构级、机构子网级及单个 IP 地址时间序列）。性能最佳的模型以粗体字和绿色阴影单元格标示，性能最差的模型则用红色阴影单元格标记；颜色越浅表示性能越优。

best or worst performance, providing a relative sense of each model's accuracy. If the values are very close or exactly the same within a row, it has the same shade.

最佳或最差性能值，用于反映各模型准确性的相对水平。若同一行内的数值非常接近或完全相同，则显示相同的色阶。

Different evaluation metrics capture distinct aspects of forecasting quality. The resulting RMSE indicates that models such as GRU, LSTM, GRU-FCN, and LSTM-FCN perform similarly well with very close results in all scenarios. GRU and LSTM models slightly outperform their FCN-enhanced counterparts in R^2 -score better-matching the temporal trend of the data.

不同的评估指标能够反映预测质量的不同维度。最终 RMSE 表明，GRU、LSTM、GRU-FCN 和 LSTM-FCN 等模型在所有场景下均表现相当优异且结果高度接近；其中 GRU 和 LSTM 模型在 R^2 评分上略优于其 FCN 增强版本，更能准确捕捉数据的时间趋势。

It can be seen that the results of the model performances depend on the window settings. For example, settings with a training window equal to or less than 168 values suggest that GRU-FCN and LSTM-FCN are quite effective in capturing overall trends in the data. Contrarily, the setting with a training window larger than 168 values suggests the opposite statement. The Mean and SARIMA do not perform well in the RMSE metric due to high prediction errors.

可见，模型性能的结果取决于窗口设置。例如，当训练窗口大小等于或小于 168 个数值时，GRU-FCN 和 LSTM-FCN 在捕捉数据整体趋势方面表现优异；反之，当训练窗口大于 168 个数值时，则效果相反。Mean 和 SARIMA 在 RMSE 指标上的表现不佳，主要源于较高的预测误差。

Mean model excels in terms of R^2 -score by accurately reflecting the general trend, although it lacks the capacity to model seasonal or cyclic variations. Among the deep learning models, RCLSTM demonstrates superior performance for one-step-ahead predictions of Institutional time series on R^2 -score. However, for the rest of the cases, it remains below average compared to the other models despite its promising superior performance described in [45]. The discrepancy is probably caused by different tested time series

均值模型在 R^2 分数方面表现优异，能够准确反映总体趋势，但缺乏建模季节性或周期性变化的能力。在深度学习模型中，RCLSTM 在机构时间序列的一步预测 R^2 分数上表现出卓越性能；然而在其余场景中，尽管其性能如[45]所述具有显著优势，仍低于其他模型的平均水平。这种差异很可能源于所测试时间序列的不同。

³<https://github.com/koumajos/isp-forecasting-benchmark>

TABLE IV
表 IV

OVERALL MEAN RMSE SHOWS THAT BASELINE MEAN AND SARIMA YIELD THE LARGEST ERRORS (LOWER IS BETTER), WHILE GRU, LSTM, GRU-FCN, AND LSTM-FCN PERFORM BEST. OTHER, MORE COMPLEX MODELS SHOW HIGHER ERRORS, WITH INCEPTIONTIME EVEN UNDERPERFORMING THE BASELINE IN SEVERAL CASES

总体平均 RMSE 表明，基线模型与 SARIMA 模型产生的误差最大（数值越低越好），而GRU、LSTM、GRU-FCN及LSTM-FCN模型表现最佳；其他更复杂的模型则产生更高误差，在某些情况下 INCEPTIONTIME 甚至 UNDERPERFORMING 于基线模型。

	Training w.	Prediction w.	Mean	SARIMA	GRU	LSTM	GRU-FCN	LSTM-FCN	InceptionTime	ResNet	RCLSTM	ETSformer
Inst.	24	1	0.146	0.181	0.104	0.105	0.102	0.102	0.112	0.106	0.140	0.130
	168	1	0.149	0.144	0.105	0.106	0.104	0.103	0.124	0.127	0.140	0.129
	168	24	0.150	0.211	0.123	0.124	0.115	0.117	0.483	0.131	0.163	0.158
	744	1	0.151	0.135	0.106	0.106	0.111	0.110	0.136	0.146	0.112	0.112
	744	168	0.152	0.294	0.165	0.165	0.165	0.164	0.519	0.173	0.193	0.178
Mean			0.150	0.193	0.121	0.121	0.119	0.119	0.275	0.1366	0.149	0.144
Inst. subnets	24	1	0.383	0.407	0.218	0.219	0.217	0.217	0.230	0.220	0.323	0.338
	168	1	0.392	0.368	0.219	0.220	0.219	0.219	0.238	0.235	0.277	0.336
	168	24	0.394	0.447	0.237	0.237	0.229	0.231	0.590	0.244	0.334	0.373
	744	1	0.400	0.355	0.220	0.220	0.228	0.228	0.253	0.255	0.225	0.336
	744	168	0.402	0.574	0.265	0.265	0.269	0.265	0.613	0.275	0.326	0.399
Mean			0.394	0.430	0.231	0.232	0.232	0.232	0.384	0.245	0.297	0.356
IP addr.	24	1	1.010	1.127	0.149	0.150	0.150	0.151	0.165	0.152	0.221	0.958
	168	1	1.011	1.083	0.150	0.151	0.152	0.152	0.168	0.158	0.226	0.952
	168	24	1.011	1.197	0.154	0.154	0.154	0.158	0.573	0.159	0.247	0.976
	744	1	1.013	1.052	0.150	0.151	0.153	0.154	0.183	0.163	0.166	0.952
	744	168	1.014	1.090	0.179	0.179	0.214	0.189	0.593	0.182	0.234	0.980
Mean			1.010	1.109	0.156	0.157	0.165	0.161	0.336	0.163	0.219	0.964
Overall mean			0.518	0.578	0.169	0.170	0.172	0.171	0.332	0.182	0.222	0.488

TABLE V
表 V

OVERALL MEAN R^2 -score SHOWS BASELINE MEAN PERFORMS BEST, FOLLOWED BY GRU, LSTM, AND THEIR FCN VARIANTS, WHILE INCEPTIONTIME PERFORMS WORST

总体平均 R^2 分数显示：基线模型的平均性能最佳，其次是GRU、LSTM 及其 FCN 变体，而inceptiontime模型的表现最差。

	Training window	Prediction window	Mean	SARIMA	GRU	LSTM	GRU-FCN	LSTM-FCN	InceptionTime	ResNet	RCLSTM	ETSformer
Inst.	24	1	0.093	-0.882	0.081	0.091	0.146	0.186	-0.773	0.058	0.161	-0.087
	168	1	0.040	0.023	0.030	0.101	0.167	0.144	-1.085	-0.630	0.159	0.062
	168	24	0.028	-1.022	-0.447	-0.451	-0.181	-0.285	-9.221	-0.733	-0.406	-0.602
	744	1	0.009	0.207	0.058	0.066	-0.046	-0.086	-1.586	-0.918	0.231	0.086
	744	168	-0.008	-2.668	-0.529	-0.493	-1.082	-0.541	-9.178	-0.737	-1.800	-1.280
Mean			0.032	-0.868	-0.161	-0.137	-0.199	-0.116	-4.369	-0.592	-0.331	-0.364
Inst. subnets	24	1	0.115	-1.130	-0.099	-0.046	0.063	0.079	-1.288	-0.150	0.062	-0.396
	168	1	0.062	-0.152	-0.098	-0.070	0.053	-0.013	-1.572	-0.678	0.073	-0.141
	168	24	0.047	-1.275	-0.548	-0.535	-0.268	-0.548	-8.903	-0.775	-0.657	-0.881
	744	1	0.021	0.126	-0.104	-0.087	-0.190	-0.236	-2.192	-1.054	0.104	-0.057
	744	168	-0.001	-3.453	-0.606	-0.551	-1.674	-0.938	-8.846	-0.915	-2.256	-1.530
Mean			0.049	-1.177	-0.291	-0.258	-0.423	-0.331	-4.560	-0.714	-0.535	-0.601
IP addr.	24	1	-0.001	-3.645	-0.464	-0.410	-0.124	-0.439	-2.698	-0.808	-0.089	-1.765
	168	1	0.013	-1.644	-0.409	-0.400	-0.177	-0.664	-2.818	-1.032	-0.118	-1.199
	168	24	0.009	-2.056	-0.308	-0.348	-0.612	-1.277	-9.223	-0.916	-0.988	-1.982
	744	1	-0.078	-0.618	-0.380	-0.393	-0.254	-0.623	-3.568	-1.167	-0.285	-0.846
	744	168	-0.087	-0.884	-0.505	-0.337	-3.968	-1.524	-9.169	-0.524	-2.775	-2.185
Mean			-0.029	-1.769	-0.413	-0.378	-1.027	-0.905	-5.495	-0.889	-0.851	-1.596
Overall mean			0.004	-1.271	-0.288	-0.258	-0.550	-0.451	-4.808	-0.732	-0.572	-0.974

of the original work and the time series from real-world network traffic. Finally, no single model consistently outperforms the others when predicting over longer horizons. The ETSformer, the most complex model based on transformers, achieves comparable performance with the leading models.

该模型融合了原始数据与真实网络流量的时间序列数据。最终，在长期预测任务中，没有任何单一模型能持续超越其他模型的表现。基于Transformer架构的最复杂模型 ETSFopmer，其性能与领先模型相当。

The SARIMA model is frequently used in the network traffic domain [10], [13], [54]. In our experiments, however, it shows weak performance, with high RMSE and low R^2 -scores. This outcome is largely due to our setup, which seeks a single best model configuration applicable across all time series in the dataset. SARIMA performs well only

SARIMA 模型在网络流量领域被广泛使用[10]。[13], [54]然而在我们的实验中，该模型表现较差，RMSE 值较高而 R^2 值较低。这一结果主要源于我们的实验设置——即试图为数据集中的所有时间序列寻找统一的最佳模型配置。SARIMA 仅在特定条件下表现良好。

TABLE VI
表 VI
RESULTING MEAN *Harmonic Score* SHOWS THAT GRU, LSTM, AND THEIR FCN VARIANTS PERFORM BEST, WHILE MORE COMPLEX MODELS PERFORM WORSE
最终计算出的平均谐波分数表明：GRU、LSTM 及其 FCN 变体表现最佳，而更复杂的模型则表现较差。

	Training window	Prediction window	Mean	SARIMA	GRU	LSTM	GRU-FCN	LSTM-FCN	InceptionTime	ResNet	RCLSTM	ETSformer
Inst.	24	1	0.146	0.213	0.120	0.121	0.115	0.116	0.134	0.122	0.119	0.128
	168	1	0.152	0.151	0.122	0.122	0.117	0.118	0.157	0.162	0.118	0.122
	168	24	0.153	0.243	0.156	0.157	0.141	0.144	0.850	0.171	0.162	0.170
	744	1	0.155	0.125	0.125	0.124	0.132	0.133	0.182	0.201	0.120	0.122
	744	168	0.156	0.392	0.171	0.171	0.171	0.169	0.855	0.186	0.224	0.206
Mean 平均：平均值			0.152	0.225	0.139	0.139	0.135	0.136	0.436	0.169	0.149	0.150
Inst. subnets	24	1	0.177	0.292	0.152	0.154	0.148	0.150	0.174	0.155	0.180	0.155
	168	1	0.187	0.212	0.154	0.155	0.153	0.154	0.191	0.183	0.160	0.148
	168	24	0.189	0.291	0.188	0.188	0.173	0.177	0.868	0.202	0.219	0.199
	744	1	0.197	0.170	0.156	0.156	0.170	0.17	0.220	0.223	0.152	0.147
	744	168	0.202	0.526	0.203	0.201	0.210	0.202	0.872	0.222	0.270	0.244
Mean 平均：平均值			0.190	0.298	0.171	0.171	0.171	0.170	0.465	0.197	0.196	0.179
IP addr.	24	1	0.300	0.852	0.129	0.131	0.131	0.132	0.159	0.134	0.164	0.151
	168	1	0.297	0.662	0.130	0.131	0.133	0.135	0.165	0.145	0.162	0.140
	168	24	0.298	0.702	0.138	0.138	0.137	0.146	0.938	0.148	0.175	0.189
	744	1	0.301	0.482	0.130	0.131	0.136	0.137	0.196	0.155	0.138	0.139
	744	168	0.303	0.424	0.151	0.150	0.221	0.170	0.943	0.156	0.193	0.200
Mean 平均：平均值			0.300	0.624	0.136	0.136	0.152	0.144	0.480	0.148	0.166	0.164
Overall mean 总平均数			0.214	0.382	0.148	0.149	0.152	0.150	0.460	0.171	0.170	0.163

when its hyperparameters are tuned for each individual series. However, this requirement is often impractical in real-world deployments [50]. Its performance also deteriorates with increasing proportions of zero values caused by device inactivity (see [11, Fig. 6]). In contrast, SARIMA performs reliably on time series without zero values, where it can predict the next data point based on longer historical patterns. Nevertheless, we utilize SARIMA only as a baseline for comparison in this work.

当其超参数需针对每个独立时间序列进行调参时，这一要求在实际应用中往往难以实现[50]。此外，随着设备闲置导致零值比例增加，其性能也会随之下降（参见[11，图6）。相比之下，SARIMA 在不含零值的时间序列上表现稳定，能够基于更长的历史数据模式预测下一个数据点。不过，在本研究中我们仅将 SARIMA 作为基准模型用于比较。

Table VI summarizes the forecasting performance using the Harmonic score, a metric that combines multiple evaluation aspects into a single measure. The results show that the GRU, LSTM, and LSTM-FCN architectures produce very similar results in several configurations. Despite the GRU-FCN model demonstrating superior performance in most experimental settings with a training period of 168 or less, it shows a significant drop in the score in the setting involving the IP address dataset, with a training window of 744 and a prediction window of 168. This decrease prevented GRU-FCN from outperforming other DL models in the overall mean.

表VI总结了使用谐波分数（一种将多个评估维度整合为单一指标的度量方法）得出的预测性能结果。数据显示，在多种配置下，GRU、LSTM 和 LSTM - FCN 架构产生的预测效果高度相似。尽管GRU- FCN 模型在训练周期不超过168小时的大多数实验场景中表现优异，但在涉及IP地址数据集、训练窗口为744小时且预测窗口为168小时的场景中，其得分显著下降。这一下降导致GRU- FCN 模型的整体平均性能未能超越其他深度学习模型。

Overall, classical architectures such as GRU, LSTM, and their convolutionally enhanced variants achieve similarly accurate results in terms of RMSE, R^2 -score, and Harmonic score. On the macro average, GRU-FCN performs best for the Institution dataset, LSTM-FCN for the Institution subnets dataset, and GRU for the IP address dataset. Considering the overall macro average across all datasets, GRU produces the most precise forecasts, followed by LSTM.

总体而言，GRU、LSTM 及其卷积增强版本等经典架构在 RMSE 值、 R^2 分数和谐波分数方面均能取得相近的准确性能。从宏观平均值来看：GRU- FCN 在机构数据集表现最佳，LSTM - FCN 在机构子网数据集表现最优，而GRU在IP地址数据集表现最为出色。综合所有数据集的宏观平均结果，GRU可提供最精准的预测结果，其次是LSTM。

The transformer-based model ETSformer achieves an overall macro average result close to leading models. Mainly, the results below average on the IP address dataset part, causing the difference between ETSformer and the leading models.

基于变压器的模型ETSformer在整体宏观平均性能上接近领先模型水平。但在IP地址数据集部分，其表现明显低于平均水平，这导致ETSformer与领先模型之间存在显著差异。

We also observe that InceptionTime is the weakest performer across all evaluation measures, followed by ResNet. InceptionTime, originally designed for time series classification, struggles with network traffic forecasting due to its reliance on multi-scale convolutions, which increase model complexity without yielding significant performance gains. Its limited ability to capture long-term dependencies further hampers its accuracy. Similarly, ResNet, despite the use of residual connections, appears less effective in modeling sequential patterns, potentially leading to suboptimal generalization at high computational costs. These findings highlight the need for architectures better suited to network time series forecasting, such as those incorporating recurrent or attention-based mechanisms.

我们还发现，在所有评估指标中，InceptionTime的表现最为逊色，其次是ResNet。InceptionTime最初是为时间序列分类设计的，但由于其依赖多尺度卷积运算（这会增加模型复杂度却无法显著提升性能），在网络流量预测任务中表现欠佳。其捕捉长期依赖关系的能力有限，进一步影响了预测精度。同样地，尽管ResNet采用了残差连接结构，但在建模序列模式方面效果较差，可能导致在高计算成本下泛化能力不足。这些研究结果凸显了需要开发更适合网络时间序列预测的架构方案，例如采用循环机制或注意力机制的模型。

B. Factors Influencing Model Predictability

B. 影响模型预测能力的因素

A clear performance trend emerges across the different dataset parts, reflecting the models’ ability to capture data temporal trends based on the R^2 -score. Forecasting performance is highest for the Institution dataset, declines for Institution subnets, and is lowest for the IP address sample. This degradation likely stems from the increasing proportion of missing values, which are minimal in the Institution dataset but significantly more prevalent in the IP address sample. The impact of missing data is further supported by the strong negative correlation (−0.69 for the IP address dataset) between the ratio of missing samples and the R^2 -score.

在不同数据集部分中均呈现出明显的性能趋势，这反映了模型基于 R^2 分数捕捉数据时间趋势的能力。预测性能在机构数据集中最高，在机构子集数据集中逐渐下降，在IP地址样本中最低。这种性能下降很可能源于缺失值比例的增加——该比例在机构数据集中极低，但在IP地址样本中却显著更高。缺失数据的影响进一步得到以下事实的支持：缺失样本比例与 R^2 分数之间存在强烈的负相关关系（IP地址数据集为−0.69）。

Analyzing the most and least predictable time series reveals that less predictable series often display sudden, abnormal spikes in values, while the most predictable series follow more regular patterns. This observation is supported by

对可预测性最高与最低的时间序列进行分析表明：可预测性较低的序列往往呈现数值的突然异常波动，而可预测性最高的序列则遵循更为规律的模式。这一观察结果得到了以下证据支持：

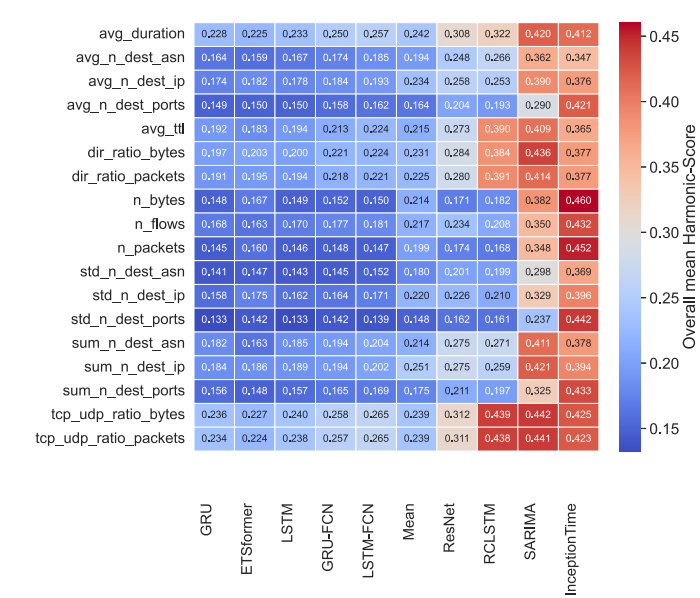


Fig. 5. Matrix providing results of overall mean *Harmonic score* for all time series metrics shows that the simpler models, including ETSformer, perform better than baseline Mean and SARIMA while leaving more complicated models behind.

图5展示了所有时间序列指标的总体平均 *谐波分数* 结果矩阵，表明包括 ETSformer 在内的简单模型表现优于基线均值和 SARIMA 模型，而更复杂的模型则表现较差。

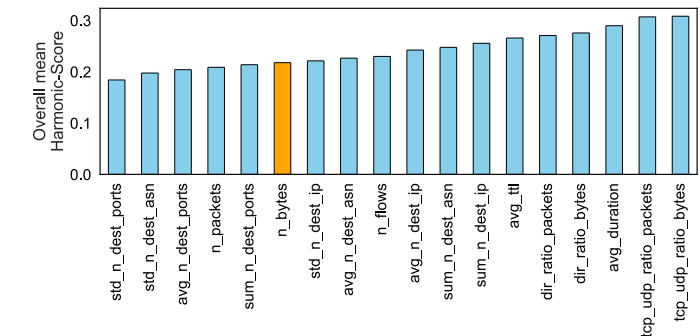


Fig. 6. Overall mean *Harmonic score* over all models for each time series metrics shows how easily predictable each time series is. The *Number of bytes* metric that is commonly utilized in network time series forecasting belongs to the less challenging ones.

图6展示了所有模型在各项时间序列指标上的总体平均 *谐波分数*，体现了各时间序列的可预测性程度。在网络时间序列预测中常用的 *字节数* 指标属于难度较低的指标之一。

autocorrelation and partial autocorrelation analyses, which show that highly predictable time series typically exhibit strong seasonal behavior.

自相关与偏自相关分析表明，高度可预测的时间序列通常表现出显著的季节性特征。

In addition, the choice of training and prediction windows significantly affects model performance. Larger prediction windows generally degrade forecast accuracy, as models struggle to capture long-term dependencies with the same precision as short-term predictions.

此外，训练窗口与预测窗口的选择会显著影响模型性能。较大的预测窗口通常会降低预测准确性，因为模型难以与短期预测相同的精度捕捉长期依赖关系。

C. Other Time Series Metrics

C. 其他时间序列指标

While the *Number of bytes* (i.e., transmitted data) is often used as a primary metric for network traffic forecasting, our results reveal that forecasting evaluation measures vary considerably across different metrics, as shown in Fig. 5 and Fig. 6. Some metrics follow similar predictability patterns to the *Number of bytes*, while others present additional challenges due to their inherent variability and dependence on specific network behaviors. Complete forecasting results for all metrics, along with code and supplementary analyses, are available on GitHub³.

虽然 *字节数*（即传输数据量）常被用作网络流量预测的主要指标，但我们的研究表明不同指标的预测评估方法存在显著差异，如图5和图6所示。部分指标的预测规律与 *字节数* 相似；而其他指标则因固有的波动性及对特定网络行为的依赖性而面临额外挑战。所有指标的完整预测结果、相关代码及补充分析均可在GitHub³上获取。

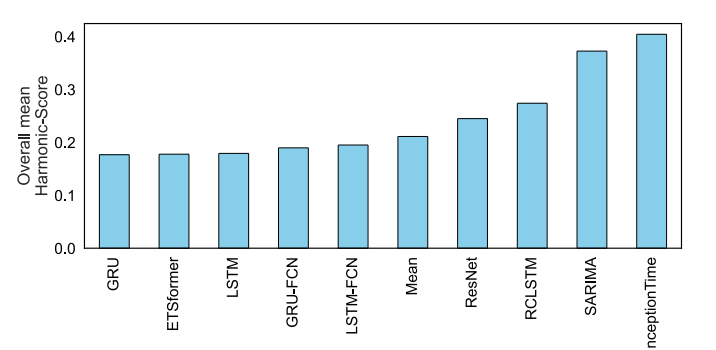


Fig. 7. Overall mean *Harmonic score* for each model shows how well the models perform among all evaluated metrics.

图7. 各模型的总体平均 *谐波分数* 反映了模型在所有评估指标中的表现水平。

Metrics representing aggregate traffic volumes, such as the *Number of flows* and *Number of packets*, generally follow similar patterns to the *Number of bytes* and are effectively captured by DL models. These metrics often exhibit seasonal behavior, making them easier to predict, especially with recurrent architectures combined with convolutional layers. The top 5 DL models perform very similarly on these metrics.

反映总流量的指标（如 *数据流数量* 和 *数据包数量*）通常与 *字节数* 呈现相似的变化趋势，且能够被深度学习模型有效捕捉。这些指标往往具有季节性特征，因此特别适合通过结合卷积层的神经网络架构进行预测。排名前五的深度学习模型在这些指标上的表现高度一致。

In contrast, forecasting more dynamic and structurally complex metrics, such as *TCP/UDP ratios* and *Direction ratios*, proves significantly more challenging. These metrics are prone to sudden fluctuations due to external influences, such as changes in user behavior. Our results show that DL models struggle to capture these fluctuations, leading to a lower R^2 -score and increased forecasting error.

相比之下，预测更具动态性且结构复杂的指标（如 *TCP/UDP 比率* 和 *方向比率*）则具有显著更高的挑战性。这些指标易受外部因素（如用户行为变化）的影响而出现剧烈波动。我们的研究表明，深度学习模型难以捕捉此类波动，导致 R^2 值降低且预测误差增大。

Overall mean result across all dataset parts and time series metrics is shown in Fig. 7. The results mostly follow the number of bytes. It can be seen that more structurally complex metrics are differentiators for the performance of models. Some models generally keep good performance regardless of the metric, while some are struggling to keep up. The ETSformer performs comparatively well and achieves the second-best position, even outcompeting other models in various metrics and measures.

所有数据集部分及时间序列指标的总体平均结果如图7所示。结果基本与字节数量呈现一致趋势：结构更复杂的指标更能有效区分不同模型的性能表现。部分模型在各类指标上均保持优异表现，而另一些模型则难以跟上竞争压力。ETSformer 模型表现尤为突出，不仅位列第二佳，还在多项指标和评估标准中超越了其他模型。

We performed a statistical significance test to objectively validate the observed performance differences between models. We used a one-way ANOVA F-test, which compares the two best models (GRU and ETSformer models). The result of the ANOVA test is shown in Table VII. Statistically significant differences ($p < 0.05$) are consistently observed across most settings, confirming the robustness of our findings. However, for one experimental setting, there is no significant difference between GRU and ETSformer.

我们进行了统计显著性检验，以客观验证不同模型之间观察到的性能差异。我们采用单因素方差分析F检验，对比了两种最佳模型（GRU模型与ETSformer模型）。方差分析结果如表VII所示。在大多数实验条件下均持续观察到统计学显著差异（ $p < 0.05$ ），这证实了我们研究结果的稳健性；然而在某一实验设置中，GRU模型与ETSformer模型之间未出现显著差异。

D. Deployability

D. 部署能力

Model deployability is a key factor; therefore, we measured the training and inference time of all DL models. The training was conducted on MetaCentrum,⁴ which manages the National Grid Infrastructure (NGI) as part of the European Grid Infrastructure (EGI). To ensure consistency, all experiments were run on an identical pool of machines with fixed configurations. Each run used a dedicated virtual environment with 4

模型的可部署性是关键因素；因此，我们测量了所有深度学习模型的训练与推理时间。训练工作在MetaCentrum平台上进行⁴，该平台作为欧洲电网基础设施（EGI）的一部分负责管理国家电网基础设施（NGI）。为确保实验一致性，所有测试均在配置固定的同一组机器上执行。每次运行均使用配备4个计算节点的专用虚拟环境。

⁴<https://metavo.metacentrum.cz/en/index.html>

TABLE VII
表 VII
RESULTS OF A ONE-WAY ANOVA F-TEST BETWEEN GRU AND
ETSFORMER MODELS WITH $\alpha = 5\%$. THE P-VALUE IS ROUNDED TO 6
DECIMAL PLACES

采用 $\alpha=5\%$ 的单因素方差分析F检验比较GRU模型与etsformer模型的结果，P值四舍五入至小数点后六位。

	Training window 培训窗口	Prediction window 预测窗口	p value p 值	Significant ($p < \alpha$)? 显著 (p < a) 吗？
Inst.	24	1	0.828959	No 否
	168	1	0.000011	Yes 是
	168	24	< 0.000001	Yes 是
	744	1	< 0.000001	Yes 是
	744	168	0.000088	Yes 是
Overall 全面的			0.0 0009	Yes 对
Inst. subnets	24	1	< 0.000001	Yes 是
	168	1	< 0.000001	Yes 是
	168	24	< 0.000001	Yes 是
	744	1	< 0.000001	Yes 是
	744	168	0.000001	Yes 是
Overall 全面的			< 0.000001	Yes 对
addr.	24	1	< 0.000001	Yes 对
			0.006204	Yes 对
	168	1		对
	168	24	< 0.000001	Yes 对
			0.000357	Yes 是
IP	744	1	< 0.000001	Yes 是
	744	168	< 0.000001	Yes 是
Overall 全面的			0.000001	Yes 对

TABLE VIII
表 VIII
RELATIVE TRAINING AND PREDICTION TIMES IN SECONDS
(NORMALIZED PER 100 DATAPOINTS)
相对训练时间与预测时间（单位：秒，按 100 数据点标准化）

	Training Inst. Sub. IP			Prediction Inst. Sub. IP		
	培训机构：I级IP子类			预测实例 I 子集 I IP		
GRU	4.24	4.50	4.2	0.11	0.12	0.14
LSTM						
激光扫描跟踪法	5.32	4.27	4.50	0.10	0.11	0.10
GRU-FCN	1.19	0.95	1.04	0.06	0.05	0.05
LSTM-FCN						
激光扫描跟踪法-FCN	3.18	3.37	3.12	0.12	0.13	0.12
InceptionTime						
《筑梦空间》上映时间	2.93	3.01	2.95	0.07	0.07	0.08
ResNet						
残差网络	6.36	5.95	5.49	0.31	0.31	0.32
RCLSTM						
残差网络	11.37	8.44	16.28	0.30	0.28	0.28
ETSformer	181.14	176.22	160.26	3.35	3.28	3.28

exclusive CPUs, minimizing interference from other processes and ensuring comparable results.

专用CPU，可最大限度减少其他进程的干扰并确保获得可比结果。

Table VIII compares the relative training and prediction times of each model, normalized to 100 data points over all experiments. These metrics highlight the computational feasibility of each model for real-world deployment. Models with long training times may be impractical for frequent retraining, while high prediction latency can hinder responsiveness, particularly when forecasting a single next value. The results show that GRU-FCN consistently achieves the shortest training and prediction times across all dataset segments and experiments, making it a strong candidate for scenarios with limited time or computational resources. Moreover, the training and prediction times of ETSformer exceeded all other models by more than ten times. Therefore, ETSformer deployability is questionable, even with overall second-best results.

表VIII比较了各模型的相对训练时间与预测时间（均以全部实验中的100个数据点为基准进行标准化）。这些指标凸显了各模型在实际应用中的计算可行性：训练耗时较长的模型可能难以满足频繁重训练的需求；而较高的预测延迟会严重影响响应速度，尤其是在预测单个后续值时。结果表明，在所有数据集片段和实验中，GRU-FCN 始终保持着最短的训练时间和预测时间，使其成为时间或计算资源有限场景下的理想选择。此外，ETSformer的训练与预测时间均超过其他所有模型十倍以上，因此即使其整体表现位列第二，其实际应用价值仍值得商榷。

VII. DISCUSSION

第七章。讨论

In this section, we discuss key findings related to the four research questions (RQs) investigated in this study. We examine factors affecting forecasting precision, compare the performance of state-of-the-art deep learning models on a public real-world network dataset, and highlight the practical implications of our results for network traffic

forecasting.

在本节中，我们讨论了本研究针对四个研究问题（RQs）所获得的关键发现。我们分析了影响预测精度的因素，比较了最先进的深度学习模型在公开真实网络数据集上的性能表现，并重点阐述了我们的研究成果对网络流量预测的实际意义。

RQ1: *What is the performance of state-of-the-art deep learning models for real-world network traffic forecasting?*

问题1: *最先进的深度学习模型在实际网络流量预测中的性能如何？*

Our evaluation shows that state-of-the-art deep learning models can achieve competitive forecasting accuracy for real-world network traffic. However, no single model consistently outperforms others across all experimental settings, including dataset partitions, training windows, and prediction windows. The results suggest that while deep learning models can provide accurate forecasts, their effectiveness depends heavily on the specific experimental conditions.

我们的评估表明，最先进的深度学习模型能够为真实网络流量提供具有竞争力的预测精度。然而，在所有实验场景（包括数据集划分、训练窗口和预测窗口）中，均未发现单一模型能持续优于其他模型。结果表明：尽管深度学习模型可提供精准预测，但其有效性在很大程度上取决于具体的实验条件。

GRU and LSTM architectures achieve the best overall results, while their hybrid variants with fully convolutional networks (GRU-FCN and LSTM-FCN) perform slightly better on the Institution and Institution subnets timeseries. Notably, GRU-FCN offers a balanced trade-off between accuracy and computational efficiency. In contrast, models like InceptionTime and ResNet underperform in most settings, possibly because their hierarchical feature extraction mechanisms are less suited for capturing the temporal dependencies and variability in network traffic data.

GRU和LSTM架构取得了最佳的整体性能，而其采用全卷积网络的混合变体（GRU-FCN和LSTM-FCN）在机构级及机构子网时间序列数据集上的表现略胜一筹。值得注意的是，GRU-FCN在准确性和计算效率之间实现了平衡。相比之下，InceptionTime和ResNet等模型在多数场景下表现欠佳，这可能是因为其分层特征提取机制不太适合捕捉网络流量数据中的时序依赖关系与变化特性。

RQ2: *How do the training and prediction windows impact the forecasting performance of state-of-the-art deep learning models?*

RQ2: *训练窗口与预测窗口如何影响最先进深度学习模型的预测性能？*

The selection of training and prediction windows significantly impacts model accuracy, balancing the trade-off between capturing long-term dependencies and managing computational overhead. Short training windows often fail to capture temporal patterns beyond immediate fluctuations, leading to suboptimal performance. Conversely, excessively long training windows introduce noise and increase computational costs without substantial accuracy gains. However, our results show that the highest precision was achieved with small training windows, while increasing the window size led to a decline in performance.

训练窗口与预测窗口的选择会显著影响模型的准确性，需要在捕捉长期依赖关系与控制计算开销之间取得平衡。过短的训练窗口往往无法捕捉超出短期波动范围的时间规律，导致模型性能欠佳；而过长的训练窗口则会引入噪声并增加计算成本，却难以显著提升准确率。然而我们的研究表明：使用较小的训练窗口时可获得最高精度，而增大窗口尺寸反而会导致性能下降。

For prediction windows, we observe a consistent trend: forecasting single next value yields significantly higher accuracy than longer prediction horizons. The degradation in accuracy over extended prediction windows is primarily due to error accumulation, a phenomenon observed across all evaluated DL models. Notably, LSTM and GRU exhibit the most stable performance across different prediction windows, highlighting their robustness in handling extended forecasts.

对于预测窗口而言，我们观察到一个一致趋势：预测单个后续值的准确率显著高于较长的预测时间范围。随着预测窗口延长，准确率下降主要源于误差累积——这一现象在所有评估的深度学习模型中均存在。值得注意的是，LSTM和GRU在不同预测窗口下均表现出最稳定的性能，凸显了其处理长周期预测时的稳健性。

Our findings suggest that much of the network traffic time series lacks strong seasonality or contains large outliers, which models with short training and prediction windows handle more effectively. As a result, larger windows perform worse in this study. However, longer training and prediction windows could be advantageous in real-world deployments, as they require fewer inferences and are less reactive to abnormal spikes, making them potentially useful for anomaly detection.

我们的研究结果表明，大部分网络流量时间序列数据缺乏明显的季节性特征或存在大量异常值；而采用较短训练窗口和预测窗口的模型能更有效地处理这类数据。因此，在本研究中，较长的窗口参数表现较差。然而，在实际应用中，较长的训练窗口和预测窗口可能更具优势——因其所需的推断次数较少，且对异常峰值反应灵敏度较低，从而有望成为异常检测的有效工具。

RQ3: *How does the granularity of aggregation (e.g., individual IP addresses, subnets, or institutions) impact the forecasting performance of state-of-the-art deep*

learning models?

问题3: 聚合粒度（例如单个IP地址、子网或机构）如何影响最先进的深度学习模型的预测性能？

The forecasting precision of DL models is strongly influenced by the level of aggregation. Our results show that the ability to capture trend patterns is highest at the institution level, decreases at the subnet level, and is lowest at the individual IP address level.

深度学习模型的预测精度受聚合层级的影响显著。我们的研究表明：捕捉趋势模式的能力在机构层面最高，在子网层面逐渐降低，而在单个IP地址层面最低。

This trend is primarily due to increasing data variability and sparsity at finer granularities. Institution-level traffic patterns are generally more stable and exhibit clear periodic trends, enabling higher predictive accuracy. In contrast, individual IP addresses often contain irregular spikes and a higher proportion of missing values, making them significantly harder to forecast. This aligns with our correlation analysis, which shows a strong negative relationship (−0.69) between missing data and R^2 -score at the IP address level. Models trained on IP-level data must contend with high noise levels and unpredictability, leading to increased forecasting errors.

这一趋势主要源于在更精细粒度层面数据变异性与稀疏性日益增加。机构层面的流量模式通常更为稳定，并呈现明显的周期性特征，从而能够实现更高的预测精度；相比之下，单个IP地址的数据往往存在不规则波动且缺失值比例较高，导致其预测难度显著增加。这与我们的相关性分析结果一致：在IP地址层面，缺失数据与 R^2 分数之间存在强烈的负相关关系（−0.69）。基于IP层面数据训练的模型必须应对较高的噪声水平和不可预测性，从而导致预测误差增大。

These findings suggest that different modeling strategies may be required for different hierarchical levels. Institution-level forecasting may be well-suited to simpler recurrent models, while IP-level forecasting may require more advanced architectures or preprocessing techniques to handle sparsity and irregularities effectively.

这些研究表明，不同层级可能需要采用不同的建模策略：机构层面的预测适合使用较为简单的循环模型；而机构内部（IP）层面的预测则可能需要更先进的架构或预处理技术，以有效应对数据稀疏性和不规则性问题。

RQ4: How do different monitored metrics (e.g., transmitted data, TCP / UDP ratio, average duration) affect the forecasting performance of state-of-the-art deep learning models?

问题4: 不同监控指标（如传输数据量、TCP/UDP比例、平均时长）如何影响先进深度学习模型的预测性能？

Forecasting accuracy varies significantly across different network traffic metrics. Metrics with clear seasonal patterns, such as the number of flows, packets, and bytes, are well predicted by DL models. In contrast, highly variable metrics, such as TCP/UDP ratios, pose challenges due to their irregular fluctuations and external dependencies, resulting in lower forecasting precision. Aggregation level also plays a crucial role—metrics at the institution level are more predictable than those at the IP level due to reduced noise. Similarly, longer aggregation windows improve performance for volume-based metrics but may obscure short-term variations in ratio-based metrics.

不同网络流量指标的预测精度存在显著差异。具有明显季节性特征的指标（如数据流数量、数据包数和字节数）能够被深度学习模型准确预测；相比之下，TCP/UDP比例这类波动性较大的指标则因波动规律不规律且存在外部依赖关系而难以预测，导致预测精度较低。聚合层级同样至关重要——由于噪声成分较少，机构层面的指标比IP层面的指标更具可预测性。类似地，较长的聚合时间窗口虽能提升基于流量的指标预测性能，但可能掩盖基于比例指标的短期波动特征。

Overall, DL models perform well on structured, high-volume metrics but struggle with fine-grained or bursty characteristics. Future research should explore adaptive strategies to improve forecasting for complex metrics, particularly assessing whether independently optimized DL models per metric outperform multivariate forecasting approaches.

总体而言，深度学习模型在结构化、高数据量的指标上表现良好，但在处理精细或突发性特征时存在困难。未来研究应探索适应性策略以提升复杂指标的预测性能，重点评估针对每个指标独立优化的深度学习模型是否优于多变量预测方法。

VIII. CONCLUSION

第八章。结论

In this study, we conducted a comprehensive evaluation of seven DL models for real-world network traffic forecasting. Our analysis revealed that LSTM and GRU surpass more complex architectures. Moreover, the GRU-FCN model achieved the best balance between predictive accuracy and computational efficiency. Beyond establishing a reproducible benchmark for network traffic forecasting using the real-world CESNET-TimeSeries24 dataset, our work lays the foundation for future work. Based on our findings, we mark several open challenges:

本研究对七种深度学习模型在实际网络流量预测中的性能进行了全面评估。分析表明，LSTM 和GRU模型的表现优于更复杂的架构；其中GRU-FCN 模型在预测精度与计算效率之间实现了最佳平衡。除利用 CESNET - TimeSeries24真实数据集建立了可复现的网络流量预测基准外，本研究还为后续研究奠定了基础。基于研究结果，我们指出了若干亟待解决的挑战：

- 1) *Multi-time-series models:* Maintaining a separate model for each time series is both computationally expensive and operationally impractical. Future work should focus on developing a universal model capable of handling different instances within the same time-series aggregation without the need for retraining.
- 1) *多时间序列模型:* 为每个时间序列维护独立模型既计算成本高昂，又在实际操作中难以实现。未来研究应重点开发一种通用模型，能够在同一时间序列聚合范围内处理不同实例而无需重新训练。
- 2) *Univariate vs. multivariate models:* Training separate models for each time series metric allows fine-tuning for specific behaviors but lacks scalability, requiring extensive computational resources and maintenance. Future work should explore the trade-offs between univariate and multivariate models for network traffic forecasting.
- 2) *单变量模型与多变量模型:* 为每个时间序列指标分别训练独立模型虽能针对特定行为进行精细调整，但缺乏可扩展性，需要消耗大量计算资源并需维护。未来研究应探讨网络流量预测中单变量模型与多变量模型之间的权衡关系。
- 3) *Handling Heterogeneous Data:* Effectively managing large-scale, heterogeneous time series with high missing value rates is crucial. While imputing gaps with zeros can degrade model performance, it remains valuable for detecting outages and other critical events.
- 3) *处理异构数据:* 有效管理具有高缺失率的大规模异构时间序列至关重要。虽然用零值填补数据缺口会降低模型性能，但这种方法对于检测系统故障及其他关键事件仍具有重要价值。
- 4) *Improving Interpretability:* Deep learning models often function as closed boxes, offering little insight into their decision-making process. Enhancing interpretability is crucial for resource allocation and anomaly detection, as it improves trust, facilitates debugging, and enables more informed decision-making.
- 4) *提升可解释性:* 深度学习模型通常像“黑箱”般运作，难以揭示其决策过程。增强可解释性对于资源分配和异常检测至关重要，因为它能提升信任度、便于调试并促进更明智的决策。

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